Fpt University

TRƯỜNG ĐẠI HỌC FPT

CAPSTONE PROJECT REPORT

Report 5 – Software Test Documentation

– Ho Chi Minh City, October 2026 –

Table of Contents

I. Record of Changes 3

II. Testing Documentation 4

1. Scope of Testing 4

2. Test Strategy 5

2.1 Testing Types 5

2.2 Test Levels 6

2.3 Supporting Tools 6

3. Test Plan 7

3.1 Human Resources 7

3.2 Test Environment 7

3.3 Test Milestones 8

4. Test Cases 8

4.1 Mandatory Features 8

4.2 Main Workflows (end to end) 16

4.3 State Transitions of the Main Entities 18

4.4 Non-Functional Requirements 22

4.5 Unit Tests 23

5. Test Reports 24

I. Record of Changes

Date	A* M, D	In charge	Change Description
28/10/2026	A	Lê Thành Minh Trung	First version for Review 2: scope, strategy, levels and tools, test plan, test case design for mandatory features, workflows, state transitions and NFRs, test report of the automated suite

*A - Added M - Modified D - Deleted

II. Testing Documentation

1. Scope of Testing

Target of test. The CampusStay API (every endpoint of `docs/api-contract.md`), the business rules behind it, the scheduled jobs, the web app and the Android app, against the functional requirements of Report 3 (FR-AUTH … FR-MOB) and the non-functional requirements NFR-01 … NFR-06.

In scope	Out of scope (reason)
The 18 mandatory features F-01 … F-18 (Design Brief section 6), each with at least one main-flow case, one data or business-rule case and one exception or permission case	Penetration testing and load above 200 concurrent users (not required for one campus; NFR-01 sets the target)
The 5 main workflows WF-01 … WF-05, end to end on the staging environment with demo data	Behaviour of Google, Firebase and Gmail themselves (stubbed or replaced in automated tests)
Every state transition of the main entities (Application, Allocation, next-Term Allocation, Invoice, payment claim, Maintenance Request) and the invalid transitions that must be refused	iOS (no iOS app)
NFR-01 … NFR-06 with their metric, threshold and method	Real bank transfers (claims are confirmed by Managers; transfers use demo accounts)
Permissions and data scope of the four roles (Student, Staff, Manager, Admin)	Browsers other than current Chrome, Edge and Firefox

Test levels.

Level	In charge	Inputs and time	Focus	Acceptance criteria
Unit	Developer of the part (Report 2, ownership)	Every change, `mvn test` (seconds)	Pure logic without the database: assignment engine, Utility split, message translation, Google code exchange, FCM sender, export quoting, CORS and mail settings	All unit tests pass
Integration	Developer of the feature; reviewed by the leader	Every change and every push, `mvn verify` locally and in GitHub Actions (about 11 minutes on a laptop)	Each endpoint through HTTP with a real PostgreSQL 17 (Testcontainers), Flyway schema and the test seed; rules, permissions, scope, jobs, events	All integration and architecture tests pass; the deploy job runs only then
System	Tester (Võ Minh Luân) with each workflow owner	Before each review and before Defense 1.1, on staging with the demo stages	End-to-end workflows and demo scenarios through the web and Android apps; NFR measurements	100 % of main scenarios and ≥ 80 % of exception scenarios pass; NFR thresholds met
Acceptance	Mentor and the team; the council at the defenses	Weekly mentor meetings; Review 2; Defense 1.1	Demo scenarios DS-01 … DS-10 on the data the council chooses	Scenarios accepted; no open Critical or High defect

Constraints and assumptions. The dormitory is not operating yet, so test data are fictional (8 Blocks, 3,680 Beds, 6,200 students in the base data; 2 Blocks and 48 students in the test seed). Time-driven rules (deadlines, milestones, jobs) are tested by moving the test clock or the dates in the data, not by waiting. Google sign-in is tested with its token check stubbed and once by hand with a real FPT account; push and email are tested against stub servers. The production server is also the staging environment until the final defense (Report 4 section 1.3.4).

2. Test Strategy

2.1 Testing Types

Type	Objective	Technique	Completion criteria
Functional testing	Every FR does what Report 3 says, for the right role	Black-box API tests per endpoint (MockMvc over HTTP with JSON), equivalence classes and boundary values from the business rules, decision tables for the assignment and the refund rules	All feature cases pass; every endpoint verified by at least one integration test (`docs/backend/02_Endpoint_Map.md`)
State transition testing	Every transition of the main entities happens only on its trigger, and invalid ones are refused	One case per transition of the state machines of Report 4 section 3.1, plus refused transitions	All TC-ST cases pass
Business-rule and data validation testing	Calculations and validations are exact	Boundary values (deadlines, stay length, percentages), field validation per field (`details.fields`), money in integer VND	Rule and validation cases pass
Security and permission testing	Roles see and change only what they may (NFR-02)	Each protected endpoint called without a token (401), with the wrong role (403) and on another student's or Block's data (404 / 403)	All permission and scope cases pass
Integration with external services	Sign-in, push, email and QR keep working when the services answer, fail or are not configured	Stub servers (`MockRestServiceServer`, a stubbed ID-token verifier), configuration switches	Unit and integration cases pass
Architecture testing	Calls go one way: controller →	Source scan (`LayeringTest`)	Passes on every build

Type	Objective	Technique	Completion criteria
	service → repository		
Regression testing	A change breaks nothing that worked	The whole automated suite runs in CI on every push; a failing test stops the deploy	Green build before every merge to `main`
System (end-to-end) testing	The main workflows work from the first actor's action to the business result	Scenario-based manual tests on staging with the demo stages (DS-01 … DS-10)	100 % main, ≥ 80 % exception scenarios
Performance testing	NFR-01	Load script with 200 virtual users against a local server with the production configuration	p95 ≤ 1 s, errors < 1 %
Usability and localisation testing	NFR-05; phone widths	Scan of every web route in Vietnamese; manual review at 375 px and 1366 px	No English left on Vietnamese screens; no horizontal scroll at 375 px

2.2 Test Levels

Type of tests	Unit	Integration	System	Acceptance
Functional	X	X	X	X
State transition		X	X	
Business rule and validation	X	X		
Security and permission		X	X	
External services	X	X	X	
Architecture	X			
Regression	X	X		
End-to-end workflows			X	X
Performance			X	
Usability and localisation			X	X

2.3 Supporting Tools

Purpose	Tool	Vendor / in-house	Version
Unit and integration tests	JUnit 5 (Jupiter), AssertJ	Open source	Spring Boot 4.1.1 managed
API tests through HTTP	Spring Boot Test, MockMvc	VMware (open source)	4.1.1

Purpose	Tool	Vendor / in-house	Version
Real database in tests	Testcontainers PostgreSQL	Open source	2.0.5 (image `postgres:17-alpine`)
Build and test run	Maven Surefire (unit), Failsafe (integration)	Apache	Maven 3.10
Continuous integration	GitHub Actions (`backend.yml`, job `verify`)	GitHub	—
Test catalog and test documents	`docs/tools/generate-catalogs.mjs`, `docs/tools/gen-test-docs.mjs`	In-house	—
Demo data for system tests	Admin › System › Demo data (stages 0, A, B, C), `backend/src/main/resources/db/demo/`	In-house	—
Load test	Node.js script with 200 virtual users	In-house (planned)	Node 22
Manual API checks	Postman collection `postman/`	Postman	—
Browsers and devices	Chrome, Edge; Android emulator (API 35) and an Android phone	Google, Microsoft	Current

3. Test Plan

3.1 Human Resources

Worker / doer	Role	Specific responsibilities
Lê Thành Minh Trung	Leader, full-stack developer	Test strategy; unit and integration tests of payments, assignment and residence; reviews every test before merge; owner of WF-01 and WF-05
Nguyễn Xuân Lộc	Backend developer	Integration tests of the Term transition (requests window, next-Term requests, reconciliation, fees); owner of WF-03
Trần Võ Quang Duy	Frontend developer (web)	Web checks of registration periods, the Term wizard, inventory screens, the Vietnamese scan; owner of WF-02
Võ Minh Luân	Mobile developer and tester	Test documentation (this report, the Test Report workbook), system tests and demo scenarios, NFR measurements, Android tests; owner of WF-04

3.2 Test Environment

Purpose	Tool	Provider	Version
Unit and integration tests (local)	Windows 10/11, JDK 21, Maven, Docker Desktop	Temurin, Docker	Java 21, Docker 29
Integration tests (CI)	GitHub-hosted runner `ubuntu-latest`, Temurin 21, Docker	GitHub	—
Test database	PostgreSQL 17 in Testcontainers with the Flyway schema and `db/test-seed`	PostgreSQL	17

Purpose	Tool	Provider	Version
System tests (staging)	Web https://campusstay-fpt-ten.vercel.app, API https://campusstay-api.eastasia.cloudapp.azure.com/api with the demo stages	Vercel, Microsoft Azure	Current `main`
Mobile	Android emulator API 35, phones with Android 8.0 or later	Google	—
Load test	Local API with the production configuration and the base data	—	—

3.3 Test Milestones

Milestone task	Start date	End date
Unit and integration tests written with each feature (Iteration 1, slices S0–S7 and v2 phases 1–8)	01/10/2026	27/10/2026
Test case design for Review 2 (features, workflows, states, NFRs)	27/10/2026	29/10/2026
Review 2	31/10/2026	31/10/2026
Concurrency test (TC-NFR-03) and load test (TC-NFR-01)	02/11/2026	15/11/2026
End-to-end workflow tests on staging, round 1 (TC-WF-01 … 05, DS-01 … DS-10)	02/11/2026	20/11/2026
Automation of the workflow tests as API scenario tests in CI; Android UI tests	09/11/2026	30/11/2026
Regression and round 2 of system tests before Defense 1.1	01/12/2026	10/12/2026

4. Test Cases

The test cases are kept in the workbook `SWP490_FA26SE260_R2_Report5_TestReport.xlsx` (one sheet per mandatory feature, plus Workflows, State transitions and NFRs) with the fields ID, feature or workflow, pre-conditions, steps, data, expected result, how it runs and the actual result. The unit tests are in `SWP490_FA26SE260_R2_Report5_UnitTest.xls`. Every automated case names the test method that runs it; the full list of the 235 automated tests is `docs/backend/04_Test_Catalog.md`, generated from the tests' Javadoc.

Source	Count	Minimum per item	Test cases designed	Automated in CI
Mandatory features (F-01 … F-18)	18	3	79	79
Main workflows, end to end (TC-WF)	5	1	5	0
State transitions of the main entities (TC-ST)	68	1	68	57
Agreed NFRs (TC-NFR)	6	1	6	2
Total			**158**	**138 (87.3 %)**

4.1 Mandatory Features

Types: *Main* = main flow; *Rule* = data or business rule; *Exception* = error, permission or data scope. *Automated test* is the test case of `docs/backend/04_Test_Catalog.md` that runs the case; its test method is in the Test Report workbook.

F-01 Sign-in (password, Google) (All; FR-AUTH-01, 02)

ID	Type	Test case	Automated test	Result
TC-F01-01	Main	Valid credentials return a token and the contract user shape.	TC-AUTH-001	Passed

ID	Type	Test case	Automated test	Result
TC-F01-02	Main	An unknown FPT Google account signs up as a student without a password; until it sets one only the setup endpoints answer; the password follows the usual rules and is set once.	TC-AUTH-038	Passed
TC-F01-03	Rule	The 5th wrong password locks the account, even for the right password.	TC-AUTH-005	Passed
TC-F01-04	Exception	No token → 401; a student calling an Admin endpoint → 403.	TC-AUTH-004	Passed

F-02 Student accounts: verification and list import (Student, Manager; FR-AUTH-04, 05)

ID	Type	Test case	Automated test	Result
TC-F02-01	Main	The sign-up form checks every field and the student code; once sent the account is Pending verification, every Manager is told, and the student cannot apply.	TC-AUTH-039	Passed
TC-F02-02	Main	A Manager verifies a pending account (the student is told) or rejects it (locked); locking signs the student out until a Manager unlocks the account; a rejected account comes back as pending.	TC-AUTH-045	Passed
TC-F02-03	Rule	The template names the columns; an upload is previewed line by line (New, Update, Unchanged, Error with the reason) and counts the students the file leaves out; nothing changes before the confirmation.	TC-AUTH-046	Passed
TC-F02-04	Main	Confirming creates and updates the accounts, verifies the waiting one that matches, gives a fictional student's code to the real student, marks real students missing from the file Not enrolled (never fictional ones), is written to the Audit Log and cannot run twice.	TC-AUTH-047	Passed
TC-F02-05	Exception	Only a verified account may apply: a pending one sees why and is refused (ADR-26).	TC-APP-022	Passed

F-03 Registration-period request and Term publication (Manager, Admin; FR-CFG-04, 05)

ID	Type	Test case	Automated test	Result
TC-F03-01	Main	A Manager files a request; it is Submitted with its proposal and history, every Admin and every other Manager is notified with a link to the requests list, and the requester is not.	TC-CFG-016	Passed
TC-F03-02	Main	The Admin builds the Draft from a Manager's request: it holds the request's dates, Blocks, Rooms and prices, the request is In progress (the Manager is told) and building again opens the same Draft; a price that differs from the request needs a reason, which the request then shows next to the difference.	TC-CFG-019	Passed
TC-F03-03	Rule	The Timeline step checks spec §5.2 per field: registration closing before it opens, renewal milestones filled in only partly or out of order, Check-in before the Term before is left.	TC-CFG-020	Passed

ID	Type	Test case	Automated test	Result
TC-F03-04	Rule	A Draft with unsaved steps cannot be published; once every step is saved, publishing fixes the Term's prices, Deposit and rules, tells every Manager (Term dashboard link) and puts the Term announcement on the students' board and bell. A later default-price change leaves the Term's prices alone, an Application pays the Term's Deposit, and once an Invoice exists the prices are locked.	TC-CFG-021	Passed
TC-F03-05	Exception	Missing fields and an unknown Block are named per field; a second open request for the same Term is refused; Staff and students cannot file or read requests.	TC-CFG-017	Passed

F-04 Apply for a Term (Student; FR-APP-01..03)

ID	Type	Test case	Automated test	Result
TC-F04-01	Main	Submit with a Deposit Invoice, then the list, detail, Home and notification; a second one is refused.	TC-APP-002	Passed
TC-F04-02	Main	An invited pair reaches the office together: the first to pay waits as Awaiting Roommate; both are Submitted when the second Deposit is paid.	TC-APP-005	Passed
TC-F04-03	Rule	The stay length (spec §6.6, D-13) is 1 to stay.max_terms, capped by the semesters the student has left (stay.program_semesters); it is kept on the Application and shown to the student and the back office; a student who already holds a Bed for the Term cannot apply (not a continuing resident).	TC-APP-023	Passed
TC-F04-04	Rule	Contact, consent and roommate rules are named per field.	TC-APP-003	Passed
TC-F04-05	Exception	Staff cannot use student screens; another student's Application is 404.	TC-APP-009	Passed

F-05 VietQR payment claim and confirmation (Student, Manager; FR-BIL-01..03)

ID	Type	Test case	Automated test	Result
TC-F05-01	Main	The claim waits for a Manager and stops the Deposit window (the Application does not expire); the queue lists it with the count and total; a bulk confirmation pays each Invoice at the time of its claim, one audit entry each, and the Applications are submitted; unknown Invoices are skipped.	TC-BIL-032	Passed
TC-F05-02	Rule	A rejection needs a reason; the Invoice is Unpaid again and the deadline continues with the time that was left (at least payment.rejected_claim_min_hours); the student is told; a second rejection cancels the Deposit and the Application expires.	TC-BIL-033	Passed
TC-F05-03	Exception	One claim at a time; a paid or void Invoice and a Deposit past its deadline cannot be claimed; a student cannot cancel while a claim waits; only Managers see the queue and decide; an	TC-BIL-034	Passed

ID	Type	Test case	Automated test	Result
		empty selection is refused; Invoices without a claim are skipped.		

F-06 Review and assignment (Manager; FR-APP-05, FR-ASG-01..03)

ID	Type	Test case	Automated test	Result
TC-F06-01	Main	The Manager approves in bulk, runs, resolves the conflict to the waitlist and commits the Block, which approves it at once (ADR-19): Allocations, Balance Invoices, notifications and My Room. Staff cannot commit.	TC-ASG-010	Passed
TC-F06-02	Rule	The queue, the review view (opening it starts the review), rejection needs a note, once only.	TC-APP-010	Passed
TC-F06-03	Rule	The priority group goes first; the same input gives the same proposal.	TC-ASG-005	Passed
TC-F06-04	Exception	A dry run is a preview; students and Staff cannot use runs (ADR-19).	TC-ASG-011	Passed
TC-F06-05	Exception	Every Manager sees every Block's Applications (ADR-19); Staff and students cannot review.	TC-APP-011	Passed

F-07 Balance, forfeit and door code release (Student, Staff, System; FR-ASG-04, 11, FR-INV-04)

ID	Type	Test case	Automated test	Result
TC-F07-01	Main	Assigned and unpaid (yellow); paid but before the check-in start, then without a Room code, the student waits (no code); Staff confirm the Room's code and it is released: the student is a resident (orange), is told and sees the code; a student who pays later receives the current code at once.	TC-ASG-032	Passed
TC-F07-02	Main	Staff draw a random code, read it to set the lock, confirm it, and it becomes the Room's code; the database holds it encrypted and every view is in the Audit Log.	TC-INV-019	Passed
TC-F07-03	Main	A student whose Balance is paid and whose check-in start has come waits for the Room's code; once Staff confirm it, it is released to them (spec §6.8): they are told (never the code), see it and become a resident; without a Bed there is nothing to show.	TC-INV-023	Passed
TC-F07-04	Rule	A Balance past its deadline without a claim is forfeited: the Allocation is cancelled, the Balance void, the Application Forfeited, the Deposit kept by the rule "Balance not paid" (a 0 % rule records no refund; a 10 % rule one refund, never two) and the student told; a claim waiting or an approved late check-in still running keeps the Bed.	TC-ASG-033	Passed
TC-F07-05	Exception	Codes are 6 new digits; other Blocks and other roles are refused.	TC-INV-020	Passed

F-08 Check-in and check-out desk (Staff; FR-ASG-04, 06)

ID	Type	Test case	Automated test	Result
TC-F08-01	Main	The Balance includes the Utility Deposit; until it is paid the desk blocks check-in; then the student checks in with the hand-over items, once only.	TC-ASG-020	Passed
TC-F08-02	Main	Check-out settles the Utility Deposit: the share since the Room's last reading, the damage charge, the refund to the Manager; the Bed is free again, the figures stay as settled and the Room's door code must change.	TC-ASG-022	Passed
TC-F08-03	Rule	Check-out records the damage of each furniture item of the Room (spec §6.16): the damage charge is their sum, a damaged item becomes Broken with the reason in its history, an item of another Room is refused; the list shows the Term's check-out date and whether a leaver is past it.	TC-ASG-045	Passed
TC-F08-04	Exception	Another Block's Staff do not find the student, every Manager does (ADR-19); students and the Admin have no desk.	TC-ASG-021	Passed

F-09 Late check-in (Student, Staff, Manager; FR-ASG-05)

ID	Type	Test case	Automated test	Result
TC-F09-01	Main	The student asks online with a reason, the day and an evidence file, up to the check-in end plus late_check_in.max_days; the Managers are told; one request at a time; a Manager approves at most that day.	TC-ASG-037	Passed
TC-F09-02	Main	Staff record the request with evidence; the Block Manager approves a deadline, after which the desk lets the student check in even though the Term's window is closed.	TC-ASG-030	Passed
TC-F09-03	Rule	A rejection needs a reason; every Manager sees the request (ADR-19).	TC-ASG-031	Passed

F-10 Next-Term requests (Student, Staff; FR-REQ-01..05)

ID	Type	Test case	Automated test	Result
TC-F10-01	Main	Staff of the Block approve with a free matching Bed of the next Term: the next-Term Allocation moves there (the student still lives in their Room this Term) and the next-Term fee will be that Bed's price; no Invoice or refund is made; Staff of another Block see nothing; after the next-Term fees the Beds are fixed.	TC-REQ-002	Passed
TC-F10-02	Main	A direct swap of next-Term Beds: the partner accepts, Staff check and approve, the two next-Term Allocations exchange Beds while both students stay where they live this Term.	TC-REQ-010	Passed
TC-F10-03	Main	The requests window in one list (room changes, swaps, Room Class changes, extensions), oldest first; a bulk decision decides each one in its own transaction and reports those it could not (a change still waiting for consent); a bulk rejection needs a reason; Staff of another Block see nothing.	TC-REQ-021	Passed

ID	Type	Test case	Automated test	Result
TC-F10-04	Rule	A Room Class change of a Room for the next Term: every other continuing resident must agree (one in the Stay's last Term is not asked); Staff cannot approve before; the furniture check reserves the missing items of the Block store; the Room's class changes for the next Term only, and the next-Term price follows it; a refusal cancels a change; Staff of another Block see nothing.	TC-REQ-020	Passed
TC-F10-05	Rule	An extension in the Stay's last Term is approved at once (the Bed is kept, the Stay commits more Terms); with an overdue Invoice it waits for Staff, who decide it with a note to reject; the length is checked; the end of the stay releases the next-Term Bed at once; after the request deadline nothing can be filed.	TC-REQ-019	Passed
TC-F10-06	Exception	A rejection needs a listed reason; the student sees it; students may not decide.	TC-REQ-003	Passed

F-11 Requests window, reconciliation and next-Term fees (Manager, System; FR-ASG-07, 08, FR-AUTH-08)

ID	Type	Test case	Automated test	Result
TC-F11-01	Main	M2 opens the requests window once: a resident with Terms left keeps the Bed as continuing, one in the Stay's last Term as awaiting extension; a student not living in the Room yet and a terminated resident get none. My Room shows the next Term with its price, the students are told, Staff and Managers read the overview, and the new students' assignment of the next Term waits for the request deadline.	TC-ASG-038	Passed
TC-F11-02	Main	M3 closes filing: a Bed still awaiting extension is released (the student is told in-app and by email, with the day to check out), a continuing one stays; the overview counts the ending.	TC-ASG-039	Passed
TC-F11-03	Main	The enrolled list of the next Term: a continuing resident who graduated (group 1, the directory updated: group 4), a continuing resident missing from the list (group 2), an unknown code (group 3), another student's changed status (group 4); the template; Managers upload, the Admin reads, Staff may not; a file without a student code column is refused.	TC-AUTH-053	Passed
TC-F11-04	Rule	Resolving: a note is required; "Not enrolled" releases the student's Bed for the next Term (told in the app); "Enrolled" keeps everything; an item is resolved once; then nothing blocks the next-Term fees.	TC-AUTH-054	Passed
TC-F11-05	Rule	Issue next-Term fees: the preview names what blocks them (filing still open, no reconciliation); once filing closed and the enrolled list is reconciled a Manager issues one fee per continuing resident = 100 % of the next Term's price of the Bed (no Deposit); issuing again charges nobody twice; leavers receive the end-of-stay notice; Staff may not issue; the fee	TC-ASG-040	Passed

ID	Type	Test case	Automated test	Result
		cannot be claimed while Utility is overdue; paid, the Stay counts the next Term.		
TC-F11-06	Exception	M7: a next-Term fee unpaid at its deadline with no claim waiting is void and its Bed released (nothing kept); a fee whose claim waits for a Manager keeps the Bed.	TC-ASG-041	Passed

F-12 Turnover and operational moves (Staff, Manager; FR-ASG-10)

ID	Type	Test case	Automated test	Result
TC-F12-01	Main	The turnover: from the next Term's check-in start a paid continuing resident receives the Room's code and lives in the next Term (the current Term's Allocation ends); a mover's old Room needs a new code; the turnover list shows leavers, movers and the Rooms whose residents change, and Staff record their meter readings.	TC-ASG-042	Passed
TC-F12-02	Main	Staff move a resident to another Room of their Block for a reason: both meters are read, the old Room's code must change, the next-Term Bed follows, nothing is charged; the student is told; the moves list shows it; validation, other Blocks and students are refused.	TC-ASG-043	Passed
TC-F12-03	Exception	After the next-Term fees are issued, Staff may no longer move a next-Term Bed; a Manager moves an unpaid one and its fee is void and issued again for the new Bed; a paid fee allows only a Bed of the same price.	TC-ASG-044	Passed

F-13 Maintenance with routing, time limits and escalation (Student, Staff, Manager; FR-MNT-01, 02, FR-REQ-06, 07)

ID	Type	Test case	Automated test	Result
TC-F13-01	Main	A checked-in student reports with photos and follows the request; the form is validated.	TC-MNT-001	Passed
TC-F13-02	Main	Staff of the Block assign, plan the visit, post updates and resolve; the student is notified, reopens once and confirms the fix. Photos open through signed links.	TC-MNT-002	Passed
TC-F13-03	Rule	A new Maintenance Request gets its SLA deadline from the Block's timer (urgent sooner than normal) and may name a furniture item of the Room, which the queue and the detail show; the item's history keeps the repair, reported and resolved; an item of another Room is refused.	TC-MNT-008	Passed
TC-F13-04	Rule	An unassigned request escalates, then passes its timer; the Manager reassigns it once.	TC-REQ-016	Passed
TC-F13-05	Exception	The queue is scoped: another Block's Staff neither see nor count the request; Managers do (ADR-19).	TC-MNT-003	Passed

F-14 Meter readings and Utility Invoices (Staff; FR-BIL-04)

ID	Type	Test case	Automated test	Result
TC-F14-01	Main	Closing the month issues each resident's Utility Invoice with its breakdown; then it is read-only.	TC-BIL-026	Passed
TC-F14-02	Rule	Within a segment, shares follow the time each resident lived in the Room; totals round up.	TC-BIL-030	Passed
TC-F14-03	Rule	A reading may not go back; an unusual jump waits for a confirmation.	TC-BIL-024	Passed
TC-F14-04	Exception	A CSV import checks every line before it saves any.	TC-BIL-025	Passed

F-15 Utility debt and termination (System, Manager; FR-BIL-05, FR-ASG-09)

ID	Type	Test case	Automated test	Result
TC-F15-01	Main	An overdue Utility Invoice gives warning 1 (in-app and email), once; a second overdue month gives the final warning with the termination day (today + 3); paying everything closes both.	TC-BIL-037	Passed
TC-F15-02	Main	A Manager terminates a resident's stay with a reason: Notice sent with the move-out date, the student is told (in-app and email) and sees it on My Room, the desk lists the check-out; Staff read it but may not start or cancel; one termination at a time; a Manager cancels it with a reason and the student is told.	TC-ASG-035	Passed
TC-F15-03	Rule	From the termination day, two overdue months terminate the stay (notice to the student, the Managers' list); a debt reaching the Utility Deposit does too, even in its first month; with that rule off one month does not, and nothing happens before the termination day.	TC-BIL-038	Passed
TC-F15-04	Exception	Past the move-out date the termination is Overdue: Managers and the Block's Staff are told, the Room's door code must change and the student no longer sees it; the check-out closes the termination and refunds the unused room fee by its rule (Not enrolled 30 %, decided by a Manager).	TC-ASG-036	Passed

F-16 Inventory, Room Class by Term, furniture (Manager, Staff; FR-INV-01..03)

ID	Type	Test case	Automated test	Result
TC-F16-01	Main	A Manager sets Rooms' class for Spring 2027; Fall 2026 keeps the old class, the Rooms and Beds screen and the assignment inputs of Spring 2027 use the new one, conformance shows what the class lacks, and every Room's change is in the Audit Log.	TC-INV-009	Passed
TC-F16-02	Main	Staff move items inside their Block now (status follows the place, the Room's conformance changes); moving to another Block is a Manager's transfer, planned and then done by Staff; a planned move can also be cancelled.	TC-INV-017	Passed

ID	Type	Test case	Automated test	Result
TC-F16-03	Rule	A running Term keeps its classes, and a Room whose Beds are already given for the Term (an Allocation) keeps its class there, so no price already charged changes.	TC-INV-010	Passed
TC-F16-04	Exception	Only Managers set classes; students cannot read; bad input is named.	TC-INV-011	Passed

F-17 Notifications, announcements and push (All; FR-NTF-01..03)

ID	Type	Test case	Automated test	Result
TC-F17-01	Main	The Admin writes a pinned announcement for all students; a student reads it at the top of the board and in the bell, unread; marking it read in the bell (id `ann-…`) updates both; the Admin sees one read.	TC-NTF-041	Passed
TC-F17-02	Main	A notification is pushed to the user's devices after commit; an expired token is disabled.	TC-NTF-021	Passed
TC-F17-03	Exception	Staff write only to the residents of their Blocks: everybody or another Block is refused; the residents of the Block read it and other students do not; Staff end their own announcement; students cannot write one.	TC-NTF-042	Passed
TC-F17-04	Exception	Signed-in users only.	TC-NTF-003	Passed

F-18 Dashboards, reports, exports and search (Staff, Manager, Admin; FR-RPT-01..04)

ID	Type	Test case	Automated test	Result
TC-F18-01	Main	Block Manager dashboard: occupancy by floor, the SLA rate and target, what is waiting.	TC-RPT-002	Passed
TC-F18-02	Main	The dashboard of a Term shows its milestones in order with their progress, its Applications by status, the places per Room Type × Room Class × gender with the Term's prices, and the Deposits issued; Staff cannot read it and an unknown Term is 404.	TC-RPT-012	Passed
TC-F18-03	Rule	Exports are CSV files with the screen's columns, written to the Audit Log; scope and permissions.	TC-RPT-008	Passed
TC-F18-04	Exception	The back-office search finds students by code or name, Rooms with or without the dash and request codes inside the user's Blocks; Staff of another Block find neither the Room nor its resident; short queries find nothing and students may not search.	TC-RPT-013	Passed

4.2 Main Workflows (end to end)

Run on staging with the demo stages (Admin › System › Demo data), one main scenario per workflow; the exception scenarios are the demo scenarios of the Design Brief (Appendix A2).

ID	Workflow	Owner	Pre-conditions and data	Steps	Expected result	Run	Result
TC-WF-01	WF-01 Registration to residence	Lê Thành Minh Trung	Stage A (registration open); se220001 eligible, no Application	1. se220001 applies (4-bed, Standard, stay length 2) and claims the Deposit transfer. 2. manager.a confirms the claim and approves the Application. 3. Stage B: se220004 (assigned, unpaid) claims the Balance; manager.a confirms. 4. staff.a1 confirms the Room code; the release job runs.	Application Submitted then Approved; se220004 is a resident (orange Bed) and sees the code; each step notified	Manual (staging, demo data)	Not run
TC-WF-02	WF-02 Opening a registration period	Trần Võ Quang Duy	Stage 0; manager.a, admin	1. manager.a requests a period for the next Term and imports the excluded Rooms from the sample file. 2. admin builds the Term from the request, changes one price with a reason, completes the 7 steps and publishes.	Request Done and linked to the Term; manager.a sees the changed value; students see the registration card and the announcement	Manual (staging, demo data)	Not run
TC-WF-03	WF-03 Term transition of continuing residents	Nguyễn Xuân Lộc	Stage A (requests window open); se210006 in A-505	1. se210006 asks for a room change. 2. staff.a1 approves it with A-603 (stage B). 3. manager.a	The next-Term Allocation moves to A-603; fees issued once with count and total; se210001	Manual (staging, demo data)	Not run

ID	Workflow	Owner	Pre-conditions and data	Steps	Expected result	Run	Result
				uploads the enrolled list and resolves groups 1–2. 4. manager.a issues the next-Term fees; se210001 pays and claims.	is Continuing and paid		
TC-WF-04	WF-04 Maintenance with routing and time limits	Võ Minh Luân	Stage A; se200009 in A-801, routing rules of Block A	1. se200009 reports an urgent air-conditioner fault with a photo and the item. 2. staff.a2 (routed) starts and resolves it. 3. se200009 confirms.	Assigned at once with the urgent deadline; Closed; the item keeps a repair entry; notifications at each step	Manual (staging, demo data)	Not run
TC-WF-05	WF-05 Monthly Utility, debt and termination	Lê Thành Minh Trung	Stage A; staff.a1; meter sample file of Block A	1. staff.a1 imports the readings and closes the month. 2. se200001 has an overdue Utility Invoice; the overdue job runs. 3. se200001 claims the transfer; manager.a confirms.	One Utility Invoice per resident split by days; warning 1 sent once; after payment the warning closes	Manual (staging, demo data)	Not run

4.3 State Transitions of the Main Entities

One case per transition of the state machines in Report 4 section 3.1, and four invalid transitions that must be refused. *Test* is the automated test case that runs it (its test method is in the Test Report workbook and `docs/backend/04_Test_Catalog.md`); *Manual* cases have no automated test yet and are run on staging.

ID	Entity	From → To	Trigger	FR	Test	Result
TC-ST-01	Application	(new) → Awaiting Deposit	Student submits the wizard	FR-APP-02	TC-APP-002	Passed

ID	Entity	From → To	Trigger	FR	Test	Result
TC-ST-02	Application	Awaiting Deposit → Awaiting Roommate	Deposit confirmed, invited roommate not ready	FR-APP-03	TC-APP-005	Passed
TC-ST-03	Application	Awaiting Roommate → Submitted	Roommate's Deposit confirmed, or the roommate declines	FR-APP-03	TC-APP-006	Passed
TC-ST-04	Application	Awaiting Deposit → Submitted	Manager confirms the Deposit claim	FR-BIL-03	TC-BIL-032	Passed
TC-ST-05	Application	Awaiting Deposit → Expired	Deposit window ends without a claim	FR-APP-04	TC-APP-008	Passed
TC-ST-06	Application	Awaiting Deposit → Expired	Second claim rejected	FR-BIL-03	TC-BIL-033	Passed
TC-ST-07	Application	Submitted → Under Review	Manager opens the review	FR-APP-05	TC-APP-010	Passed
TC-ST-08	Application	Submitted → Approved	Manager approves (bulk)	FR-APP-05	TC-ASG-010	Passed
TC-ST-09	Application	Under Review → Approved	Manager approves after opening the review	FR-APP-05	Manual	Not run
TC-ST-10	Application	Under Review → Rejected	Manager rejects with a note	FR-APP-05	TC-APP-010	Passed
TC-ST-11	Application	Submitted → Rejected	Manager rejects from the queue without opening it	FR-APP-05	Manual	Not run
TC-ST-12	Application	Live state → Cancelled	Student cancels before the cancel deadline	FR-APP-04	TC-APP-007	Passed
TC-ST-13	Application	Live state → Rejected	Student not enrolled (leavers list)	FR-AUTH-08	TC-AUTH-055	Passed
TC-ST-14	Application	Approved → Waitlisted	No matching Bed left in the assignment	FR-ASG-03	TC-ASG-010	Passed
TC-ST-15	Application	Approved → Allocated	Assignment approved for the Block	FR-ASG-03	TC-ASG-010	Passed
TC-ST-16	Application	Waitlisted → Allocated	Round-2 offer accepted	FR-APP-07	TC-APP-020	Passed
TC-ST-17	Application	Waitlisted → Cancelled	Round-2 offer declined or expired	FR-APP-07	TC-APP-021	Passed
TC-ST-18	Application	Allocated → Forfeited	Balance not paid by its deadline	FR-ASG-11	TC-ASG-033	Passed
TC-ST-19	Application	Allocated → Rejected	Not enrolled before moving in	FR-AUTH-08	Manual	Not run

ID	Entity	From → To	Trigger	FR	Test	Result
TC-ST-20	Allocation	(new) → Assigned – unpaid	Assignment approved	FR-ASG-03	TC-ASG-010	Passed
TC-ST-21	Allocation	Assigned – unpaid → Paid – waiting for the code	Balance claim confirmed	FR-BIL-03	TC-ASG-032	Passed
TC-ST-22	Allocation	Assigned – unpaid → Cancelled	Balance past its deadline (forfeit)	FR-ASG-11	TC-ASG-033	Passed
TC-ST-23	Allocation	Assigned – unpaid → Cancelled	Balance claim rejected twice	FR-BIL-03	TC-ASG-034	Passed
TC-ST-24	Allocation	Paid – waiting for the code → Cancelled	Not enrolled before moving in	FR-AUTH-08	Manual	Not run
TC-ST-25	Allocation	Paid – waiting for the code → Resident	Check-in start reached and the Room code confirmed (job)	FR-ASG-04	TC-INV-023	Passed
TC-ST-26	Allocation	Paid – waiting for the code → Resident	Checked in at the desk	FR-ASG-04	TC-ASG-020	Passed
TC-ST-27	Allocation	Resident → Ended	Checked out at the desk	FR-ASG-06	TC-ASG-022	Passed
TC-ST-28	Allocation	Resident → Ended	Moved to another Bed	FR-ASG-10	TC-ASG-043	Passed
TC-ST-29	Allocation	Resident → Ended	The next Term starts for a continuing resident	FR-ASG-10	TC-ASG-042	Passed
TC-ST-30	Next-Term Allocation	(new) → Continuing	M2 opens the window, Terms left in the Stay	FR-ASG-07	TC-ASG-038	Passed
TC-ST-31	Next-Term Allocation	(new) → Awaiting extension	M2 opens the window, last Term of the Stay	FR-ASG-07	TC-ASG-038	Passed
TC-ST-32	Next-Term Allocation	Awaiting extension → Continuing	Extension approved	FR-REQ-04	TC-REQ-019	Passed
TC-ST-33	Next-Term Allocation	Awaiting extension → Cancelled	Not extended by M3	FR-ASG-07	TC-ASG-039	Passed
TC-ST-34	Next-Term Allocation	Continuing → Continuing (other Bed)	Room change or swap approved	FR-REQ-01, 02	TC-REQ-002	Passed

ID	Entity	From → To	Trigger	FR	Test	Result
TC-ST-35	Next-Term Allocation	Continuing → Cancelled	End of stay or not enrolled	FR-REQ-04, FR-AUTH-08	TC-AUTH-054	Passed
TC-ST-36	Next-Term Allocation	Continuing → Cancelled	Next-Term fee unpaid at M7	FR-ASG-08	TC-ASG-041	Passed
TC-ST-37	Next-Term Allocation	Continuing → Resident	Fee paid and the new Term starts	FR-ASG-08	TC-ASG-042	Passed
TC-ST-38	Invoice	(new) → Unpaid	Issued (Deposit, Balance, next-Term fee, Utility)	FR-APP-02, FR-BIL-04	TC-BIL-026	Passed
TC-ST-39	Invoice	Unpaid → Awaiting confirmation	Student claims the transfer	FR-BIL-02	TC-BIL-032	Passed
TC-ST-40	Invoice	Overdue → Awaiting confirmation	Student claims an overdue debt	FR-BIL-02	TC-BIL-035	Passed
TC-ST-41	Invoice	Awaiting confirmation → Paid	Manager confirms the claim	FR-BIL-03	TC-BIL-011	Passed
TC-ST-42	Invoice	Unpaid → Paid	Manager confirms a transfer the student did not claim	FR-BIL-03	TC-BIL-011	Passed
TC-ST-43	Invoice	Overdue → Paid	Manager confirms an overdue debt the student did not claim	FR-BIL-03	Manual	Not run
TC-ST-44	Invoice	Awaiting confirmation → Unpaid	Manager rejects the claim, time left continues	FR-BIL-03	TC-BIL-033	Passed
TC-ST-45	Invoice	Awaiting confirmation → Overdue	Rejected claim of a debt already past due	FR-BIL-03	TC-BIL-035	Passed
TC-ST-46	Invoice	Awaiting confirmation → Void	Second rejection of a Deposit or Balance claim	FR-BIL-03	TC-ASG-034	Passed
TC-ST-47	Invoice	Unpaid → Overdue	Due date passes (Utility, charges)	FR-BIL-05	TC-BIL-020	Passed
TC-ST-48	Invoice	Unpaid → Void	Forfeit or cancelled Application	FR-ASG-11, FR-APP-04	TC-APP-007	Passed
TC-ST-49	Invoice	Overdue → Void	Overdue Utility settled from the Utility Deposit at check-out	FR-ASG-06	Manual	Not run
TC-ST-50	Payment claim	(new) → Claimed	Student presses "I have transferred"	FR-BIL-02	TC-BIL-032	Passed
TC-ST-51	Payment claim	Claimed → Confirmed	Manager confirms	FR-BIL-03	TC-BIL-011	Passed

ID	Entity	From → To	Trigger	FR	Test	Result
TC-ST-52	Payment claim	(new) → Confirmed	Manager confirms a transfer the student did not claim (claim in the Manager's name)	FR-BIL-03	TC-BIL-011	Passed
TC-ST-53	Payment claim	Claimed → Rejected	Manager rejects with a reason	FR-BIL-03	TC-BIL-033	Passed
TC-ST-54	Maintenance Request	(new) → Submitted	Resident reports a problem	FR-MNT-01	TC-MNT-001	Passed
TC-ST-55	Maintenance Request	Submitted → Assigned	Routing rule or Staff assigns, deadline set	FR-REQ-06	TC-REQ-014	Passed
TC-ST-56	Maintenance Request	Assigned → Submitted	Assignee removed	FR-MNT-02	Manual	Not run
TC-ST-57	Maintenance Request	Assigned → In Progress	Staff starts the work	FR-MNT-02	TC-MNT-002	Passed
TC-ST-58	Maintenance Request	In Progress → Resolved	Staff marks it fixed	FR-MNT-02	TC-MNT-002	Passed
TC-ST-59	Maintenance Request	Assigned → Resolved	Staff fixes it without starting it first	FR-MNT-02	Manual	Not run
TC-ST-60	Maintenance Request	Resolved → Closed	Reporter confirms the fix	FR-MNT-02	TC-MNT-002	Passed
TC-ST-61	Maintenance Request	Resolved → Assigned	Reporter reopens (once)	FR-MNT-02	TC-MNT-002	Passed
TC-ST-62	Maintenance Request	Submitted → Closed	Staff closes a duplicate	FR-MNT-02	Manual	Not run
TC-ST-63	Maintenance Request	Assigned → Closed	Staff closes it	FR-MNT-02	Manual	Not run
TC-ST-64	Maintenance Request	In Progress → Closed	Staff closes it	FR-MNT-02	Manual	Not run
TC-ST-65	Invoice	Paid / Void → (refused)	Claim a Paid or Void Invoice, or a Deposit past its deadline	FR-BIL-02	TC-BIL-034	Passed
TC-ST-66	Invoice	Void → (refused)	Confirm a Void Invoice	FR-BIL-03	TC-BIL-012	Passed
TC-ST-67	Invoice	Paid → (refused)	Confirm an Invoice a second time	FR-BIL-03	TC-BIL-011	Passed
TC-ST-68	Next-Term Allocation	Continuing (fees issued) → (refused)	Staff move a next-Term Bed after the fees are issued	FR-ASG-10	TC-ASG-044	Passed

4.4 Non-Functional Requirements

ID	NFR	Metric	Threshold	Method	Run	Result
TC-NFR-01	NFR-01 Performance	p95 of the 10 most used	≤ 1 s, errors < 1 %	Load script (Node, 200 virtual users, 10	Manual (script)	Not run

ID	NFR	Metric	Threshold	Method	Run	Result
		read endpoints with 200 concurrent users on the base data		minutes) against a local server with the production configuration		
TC-NFR-02	NFR-02 Security	Permission and scope tests passing	100 %	Automated: TC-AUTH-004, 016, TC-APP-009, TC-BIL-006, TC-INV-003, 007, TC-MNT-003, TC-ASG-021, TC-CFG-012 in mvn verify	Automated	Passed
TC-NFR-03	NFR-03 Integrity	Allocations of one Bed in one Term after parallel commits	0 duplicates	Concurrency test: 20 parallel requests for the same Bed, then count active Allocations per Bed and Term	Automated (planned)	Not run
TC-NFR-04	NFR-04 Availability	Share of 5-minute health checks answering UP	≥ 99 % from 01/11/2026	External uptime check of GET /api/actuator/health every 5 minutes	Manual (monitor)	Not run
TC-NFR-05	NFR-05 Usability (two languages)	API messages and web texts with a Vietnamese translation	100 %	TC-NTF-036, 037 (MessagesTest) and the scan of every web route in Vietnamese	Automated + scan	Passed
TC-NFR-06	NFR-06 Deployability	Time from push to a healthy production API; rollback time	≤ 15 min; ≤ 5 min	GitHub Actions run duration of the last 10 deployments; timed rollback with API_TAG	Manual (measure)	Not run

4.5 Unit Tests

Class under test	Test class	Tests	What is checked
`AssignmentEngine`	`AssignmentEngineTest`	6	Pairs share an empty 2-bed Room; ineligible students are never placed; a missing partner blocks the pair; best-fit Room; priority order and the same proposal for the same input; low fit is marked
`UtilitySplit`	`UtilitySplitTest`	4	Equal shares for a whole month; shares by days lived between meter cuts; rounding up; nothing to share without an end reading
`GoogleCodeExchanger`	`GoogleCodeExchangerTest`	6	Code exchange with the redirect URI; used or expired code (401); wrong

Class under test	Test class	Tests	What is checked
			secret (503); Google unreachable (503); no secret, no call; redirect flow
`FcmPushSender`	`FcmPushSenderTest`	3	Service-account JWT buys an access token; an unregistered token is reported; the JWT is RS256 with the FCM scope
`Messages`	`MessagesTest`	2	Exact texts and texts with values are translated; what has no translation stays unchanged
`EmailSender` configuration	`EmailSenderConfigTest`	3	Demo mailbox redirect of fictional accounts; real accounts unchanged; no SMTP, nothing sent or thrown
`SecurityConfig` CORS	`SecurityConfigCorsTest`	2	Allowed origins and Vercel preview pattern; other origins refused
`ExportService`	`ExportServiceTest`	1	CSV quoting, empty cells, campus times, formula neutralisation

5. Test Reports

Automated suite. Run of `mvn verify` on 28/10/2026 (Windows 10, JDK 21, Docker; the same suite runs in GitHub Actions on every push): 235 tests, 235 passed, 0 failed.

Code	Feature	Tests	Passed	Failed	Not run
AUTH	Accounts and access	55	55	0	0
CFG	Configuration and Terms	29	29	0	0
INV	Inventory	25	25	0	0
APP	Applications	14	14	0	0
ASG	Assignment, residence, Term transition	28	28	0	0
BIL	Billing and Utility	22	22	0	0
REQ	Requests, routing, time limits	16	16	0	0
MNT	Maintenance and inspections	8	8	0	0
NTF	Notifications	22	22	0	0
RPT	Reports	14	14	0	0
	Total with a test case id	**233**	**233**	**0**	**0**

Designed test cases (section 4).

Group	Test cases	Passed	Failed	Not run yet	Test coverage	Test successful coverage
Mandatory features	79	79	0	0	100.0 %	100.0 %
Main workflows	5	0	0	5	0.0 %	0.0 %
State transitions	68	57	0	11	83.8 %	83.8 %
NFRs	6	2	0	4	33.3 %	33.3 %
Total	**158**	**138**	**0**	**20**	**87.3 %**	**87.3 %**

Analysis.

- Every automated test passes (235 of 235); every endpoint of the API is verified by at least one integration test (`docs/backend/02_Endpoint_Map.md`). The suite runs on every push in GitHub Actions and a failure stops the deployment.
- 20 designed cases are not run yet: the 5 end-to-end workflow cases, 11 state transitions without an automated test and 4 NFR measurements. They are planned for the first round of system tests on staging (section 3.3) and are known gaps G2–G4 of the Design Brief.
- Automation: 138 of 158 designed cases (87.3 %) run in CI, above the 50 % required for Defense 1.1. Of the executed cases, 100 % pass (threshold 90 %), including every main-flow case of the mandatory features.
- Defects: the defects found during Iteration 1 were fixed in the change that found them, each with a regression test (Project Tracking, sheet Defects); no Critical or High defect is open.